

Data Structure Lab3 -Arrays

Exercises and Homework

java.util Methods for Arrays

fill(A, x)

copyOf(A, n)

copyOfRange(A, s, t):

toString(A)

sort(A):

binarySearch(A, x)

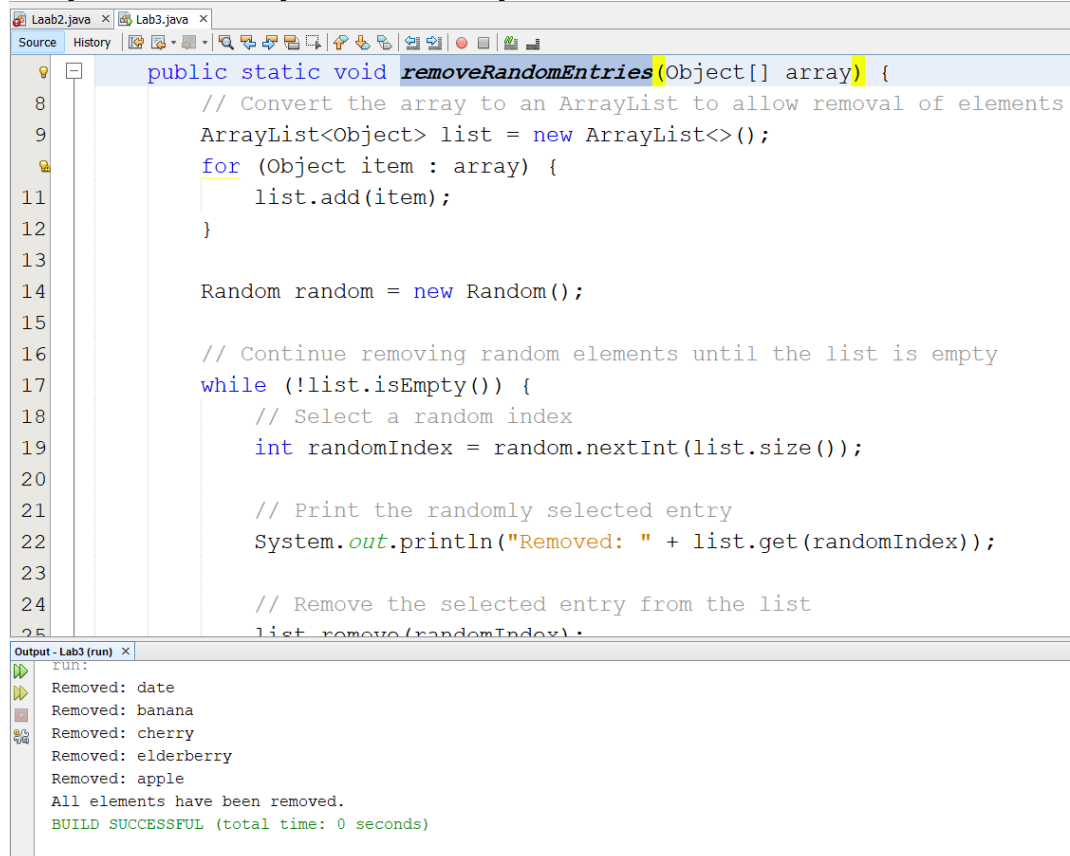
1	R-3.1	Give the next five pseudorandom numbers generated by the process described on page 113, with $a = 12$, $b = 5$, and $n = 100$, and 92 as the seed for cur. See page 113 9 13 61 37 49
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Data Structure Lab3 -Arrays

2

R-
3.2

Write a Java method that repeatedly selects and removes a random entry from an array until the array holds no more entries.

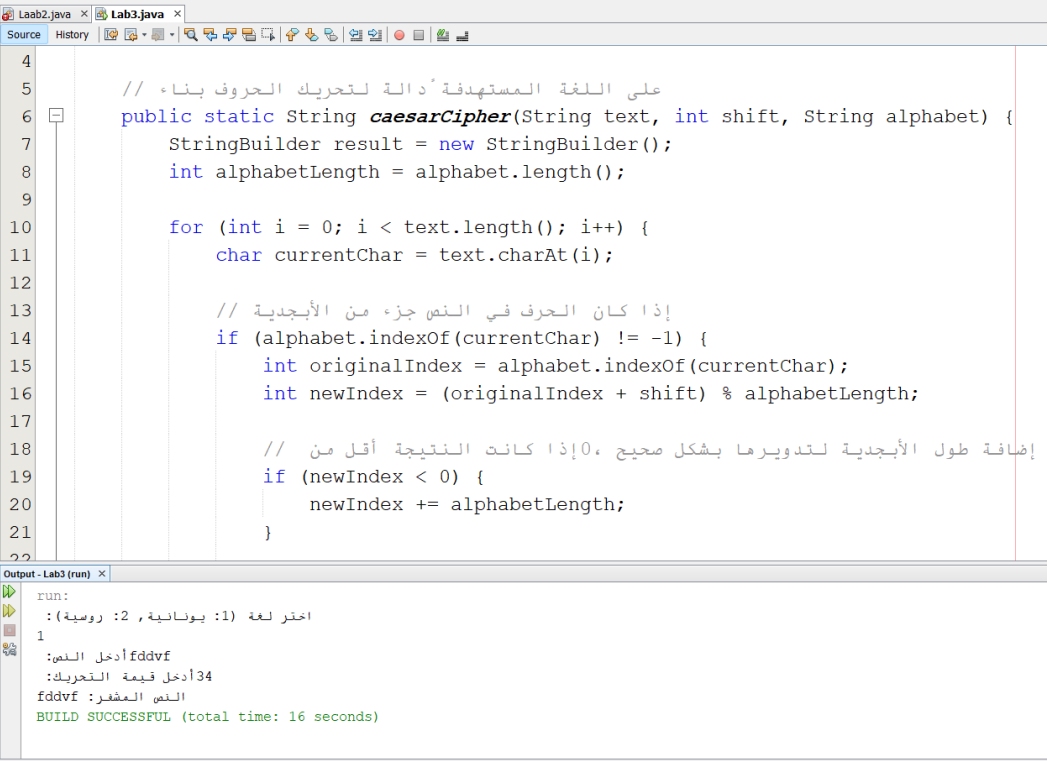


```
Laab2.java x Lab3.java x
Source History
9 public static void removeRandomEntries(Object[] array) {
8     // Convert the array to an ArrayList to allow removal of elements
9     ArrayList<Object> list = new ArrayList<>();
10    for (Object item : array) {
11        list.add(item);
12    }
13
14    Random random = new Random();
15
16    // Continue removing random elements until the list is empty
17    while (!list.isEmpty()) {
18        // Select a random index
19        int randomIndex = random.nextInt(list.size());
20
21        // Print the randomly selected entry
22        System.out.println("Removed: " + list.get(randomIndex));
23
24        // Remove the selected entry from the list
25        list.remove(randomIndex);
26    }
27 }
```

Output - Lab3 (run) x

```
Run:
Removed: date
Removed: banana
Removed: cherry
Removed: elderberry
Removed: apple
All elements have been removed.
BUILD SUCCESSFUL (total time: 0 seconds)
```

Data Structure Lab3 -Arrays

3	R-3.3	Explain the changes that would have to be made to the program of Code Fragment 3.8 so that it could perform the Caesar cipher for messages that are written in an alphabet-based language other than English, such as Greek, Russian, or Hebrew.  <pre> 4 5 // على اللغة المستهدفة دالة لتحريك الحروف بناءً 6 public static String caesarCipher(String text, int shift, String alphabet) { 7 StringBuilder result = new StringBuilder(); 8 int alphabetLength = alphabet.length(); 9 10 for (int i = 0; i < text.length(); i++) { 11 char currentChar = text.charAt(i); 12 13 // إذا كان الحرف في النص جزء من الأبجدية 14 if (alphabet.indexOf(currentChar) != -1) { 15 int originalIndex = alphabet.indexOf(currentChar); 16 int newIndex = (originalIndex + shift) % alphabetLength; 17 18 // إضافة طول الأبجدية لتدويرها بشكل صحيح، إذا كانت النتيجة أقل من 0 19 if (newIndex < 0) { 20 newIndex += alphabetLength; 21 } 22 } 23 } 24 return result.toString(); 25 } </pre> Output - Lab3 (run) x <pre> run: اختار لغة (1: يونانية, 2: روسية): 1 أدخل النص: fddvF أدخل قيمة التحريك: 34 النص المشفر: fddvF BUILD SUCCESSFUL (total time: 16 seconds) </pre>
4	R-3.4	The TicTacToe class of Code Fragments 3.9 and 3.10 has a flaw, in that it allows a player to place a mark even after the game has already been won by someone. Modify the class so that the putMark method throws an IllegalStateException in that case

Data Structure Lab3 -Arrays

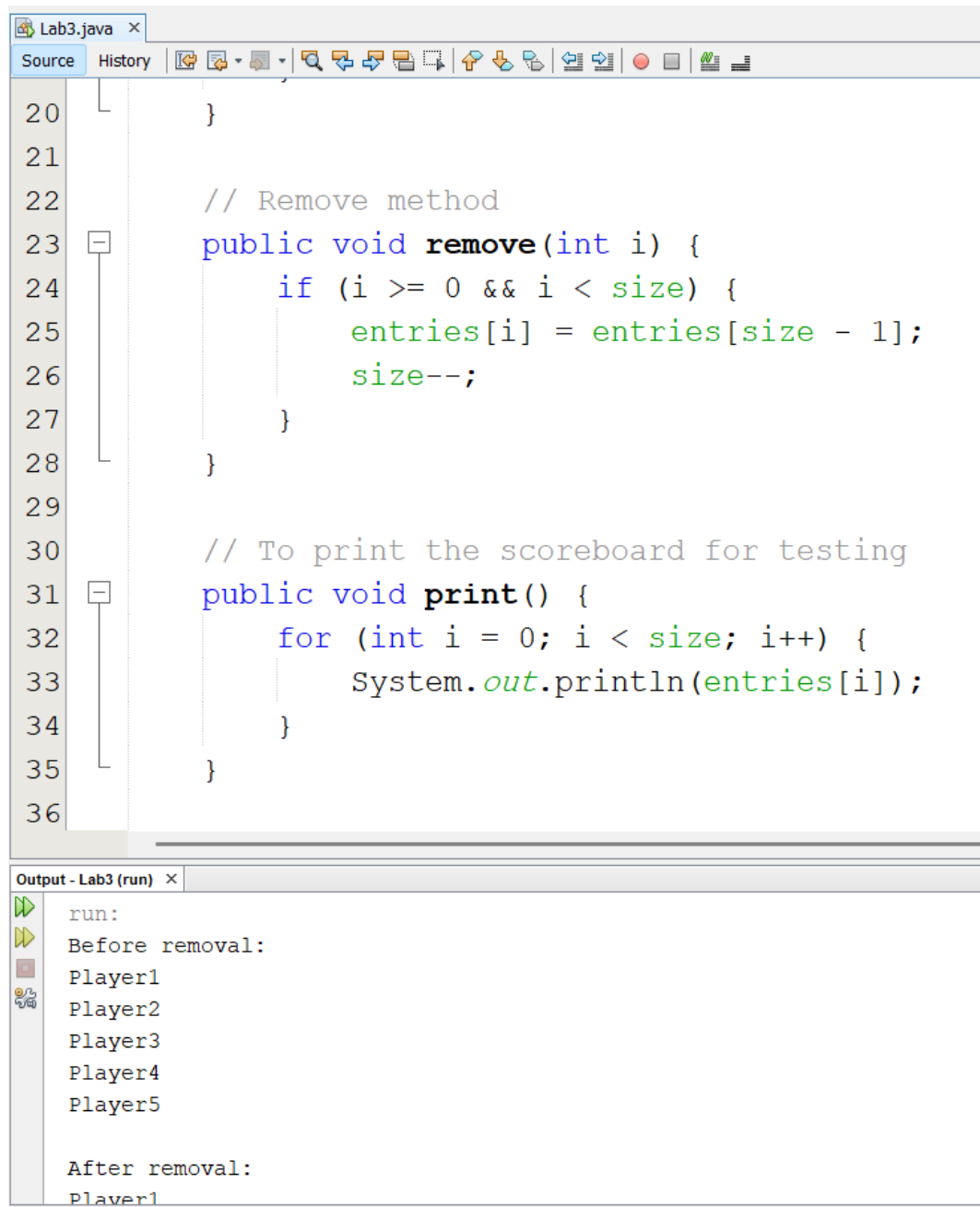
		<pre> 2 private final char[][] board = new char[3][3]; 3 private char currentPlayer = 'X'; 4 private boolean isGameOver = false; 5 6 // وضع علامة على اللوحة 7 public void putMark(int row, int col) { 8 if (isGameOver) { 9 System.out.println("لا يمكن إضافة حركة بعد انتهاء اللعبة."); 10 return; 11 } 12 if (row < 0 row >= 3 col < 0 col >= 3 board[row][col] != ' ') { 13 System.out.println("الموقع غير صالح أو مشغول بالفعل."); 14 return; 15 } 16 17 board[row][col] = currentPlayer; 18 if (checkWinner()) { 19 isGameOver = true; 20 System.out.println("اللاعب " + currentPlayer + " فاز!"); 21 } else { 22 currentPlayer = (currentPlayer == 'X') ? 'O' : 'X'; 23 } 24 } </pre>
5	R-3.1 3	What is the difference between a shallow equality test and a deep equality test between two Java arrays, A and B, if they are one-dimensional arrays of type int? What if the arrays are two-dimensional arrays of type int? (Shallow equality): أي ما إذا كانت المصفوفات تشير إلى نفس (يتم مقارنة المراجع (الكائن في الذاكرة). (Deep equality): يتم مقارنة المحتويات داخل المصفوفات نفسها، أي مقارنة العناصر نفسها داخل المصفوفات (أو المصفوفات الفرعية في حالة المصفوفات متعددة الأبعاد).
6	R-3.1 4	Give three different examples of a single Java statement that assigns variable, backup, to a new array with copies of all int entries of an existing array, original. <pre> backup = original.clone(); int[] temp = Arrays.copyOf(original, n); public static void arraycopy(Object src_array, int src_Pos, Object dest_array, int dest_Pos, int length) System.arraycopy(src_array, 0, dest_array, 0, 19); </pre>

Data Structure Lab3 -Arrays

7	C-3.17	Let A be an array of size $n \geq 2$ containing integers from 1 to $n-1$ inclusive, one of which is repeated. Describe an algorithm for finding the integer in A that is repeated. الطريقة لاكتشاف الرقم المكرر في المصفوفة هي استخدام خوارزمية اكتشاف الدورة (Floyd's Tortoise and Hare) (Cycle-Finding Algorithm).
8	C-3.18	Let B be an array of size $n \geq 6$ containing integers from 1 to $n-5$ inclusive, five of which are repeated. Describe an algorithm for finding the five integers in B that are repeated. HashSet استخدام خوارزمية تعتمد على • نقوم بالمرور عبر المصفوفة • لتخزين الأعداد التي تم رؤيتها HashSet نستخدم • ، نقوم بتخزينه في قائمة (HashSet) إذا تم العثور على عدد مكرر (موجود في الـ النتائج). • نوقف البحث عندما نجد ٥ أعداد مكررة

Data Structure Lab3 -Arrays

- 9 C-3.1 9 Give Java code for performing add(e) and remove(i) methods for the Scoreboard class, as in Code Fragments 3.3 and 3.4, except this time, don't maintain the game entries in order. Assume that we still need to keep n entries stored in indices 0 to n-1. You should be able to implement the methods without using any loops, so that the number of steps they perform does not depend on n.



The screenshot shows an IDE window titled 'Lab3.java' with a 'Source' tab. The code is as follows:

```
20     }
21
22     // Remove method
23     public void remove(int i) {
24         if (i >= 0 && i < size) {
25             entries[i] = entries[size - 1];
26             size--;
27         }
28     }
29
30     // To print the scoreboard for testing
31     public void print() {
32         for (int i = 0; i < size; i++) {
33             System.out.println(entries[i]);
34         }
35     }
36
```

Below the code editor is an 'Output - Lab3 (run)' window. It shows the following output:

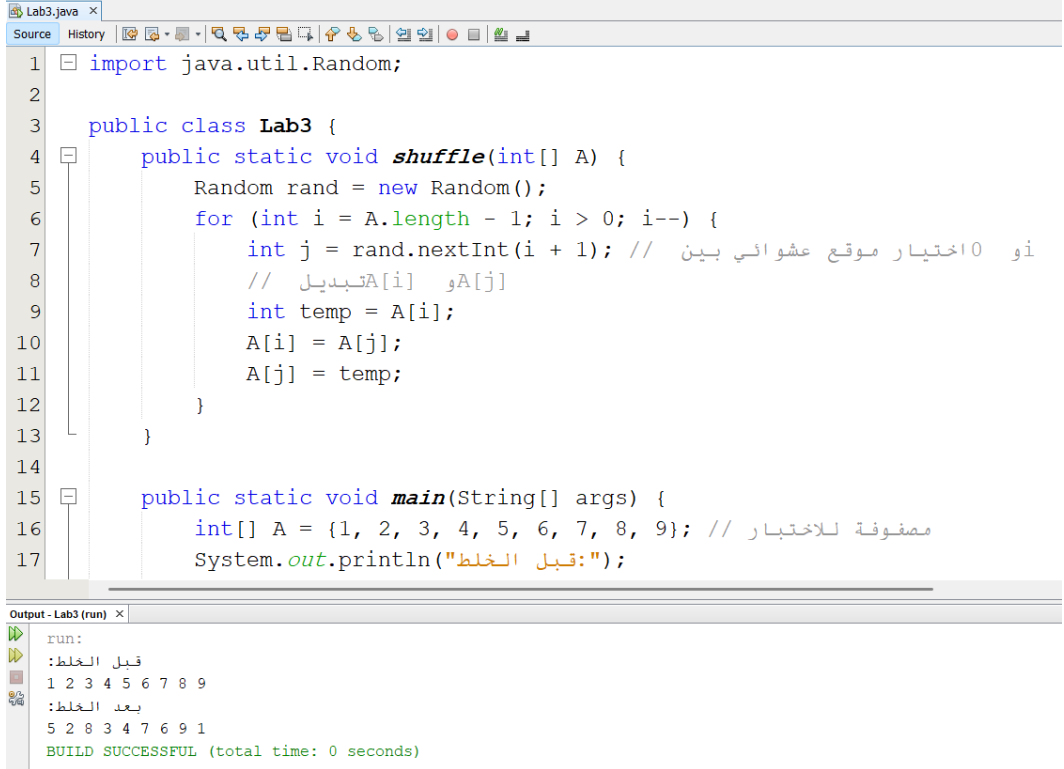
```
run:
Before removal:
Player1
Player2
Player3
Player4
Player5

After removal:
Player1
```

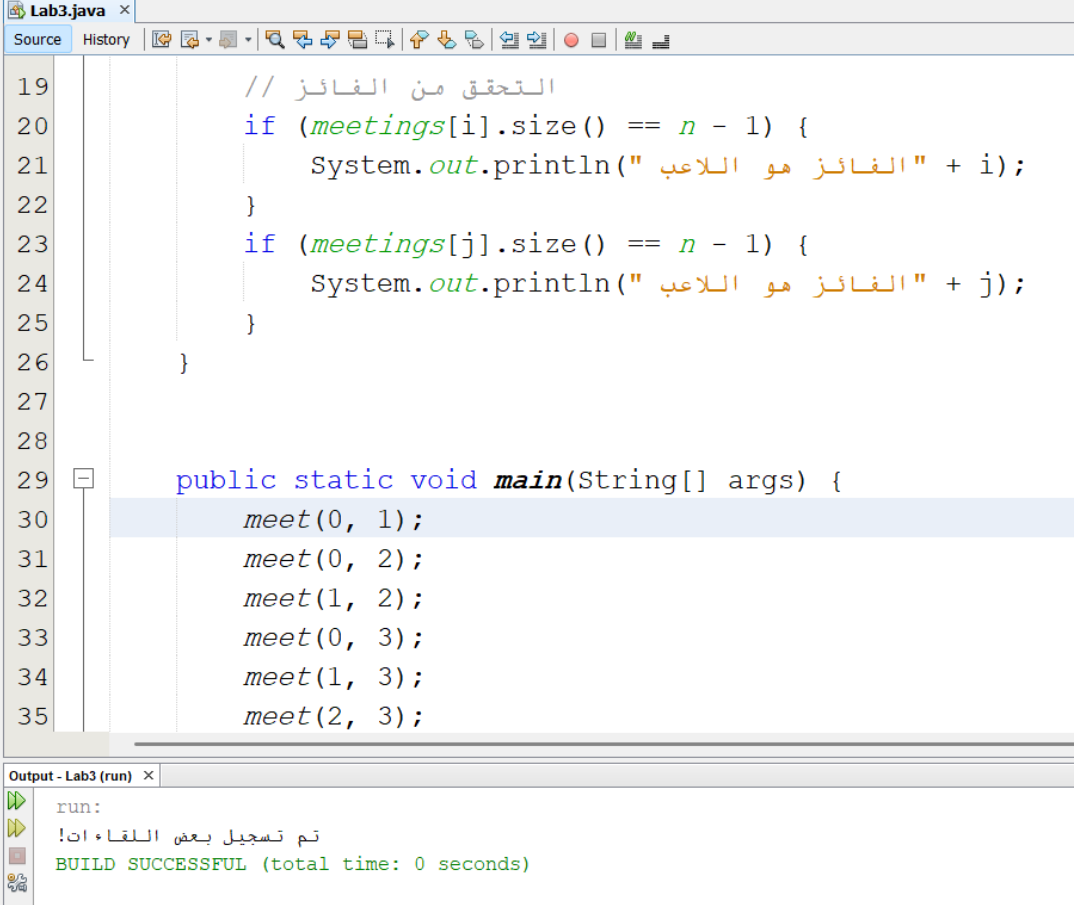
Data Structure Lab3 -Arrays

1 0	C- 3.2 0	Give examples of values for a and b in the pseudorandom generator given on page 113 of this chapter such that the result is not very random looking, for n = 1000. بحيث لا يكون التسلسل عشوائياً، يمكنك استخدام القيم التالية b و a لاختيار قيم لـ: 1. $a=1, b=0$: جميع القيم ستكون متطابقة. 2. $a=2, b=0$: التسلسل سيكون بشكل مضاعف. 3. $a=1000, b=0$: التسلسل سيتوقف عند 0 بعد أول خطوة. 4. $a=500, b=500$: سيحدث تكرار سريع في التسلسل. هذه القيم تؤدي إلى تسلسلات غير عشوائية وقابلة للتنبؤ.
1 1	C- 3.2 1	Suppose you are given an array, A, containing 100 integers that were generated using the method <code>r.nextInt(10)</code>, where r is an object of type <code>java.util.Random</code>. Let x denote the product of the integers in A. There is a single number that x will equal with probability at least 0.99. What is that number and what is a formula describing the probability that x is equal to that number? • العدد الذي سيكون x مساوياً له باحتمال لا يقل عن 0.99 هو 0. • الصيغة التي تصف الاحتمال هي: $100 \left(\frac{9}{10} \right) - 1 = 0 = P(x)$
1 2	C- 3.2 2	Write a method, <code>shuffle(A)</code>, that rearranges the elements of array A so that every possible ordering is equally likely. You may rely on the <code>nextInt(n)</code> method of the <code>java.util.Random</code> class, which returns a random number between 0 and n-1 inclusive.

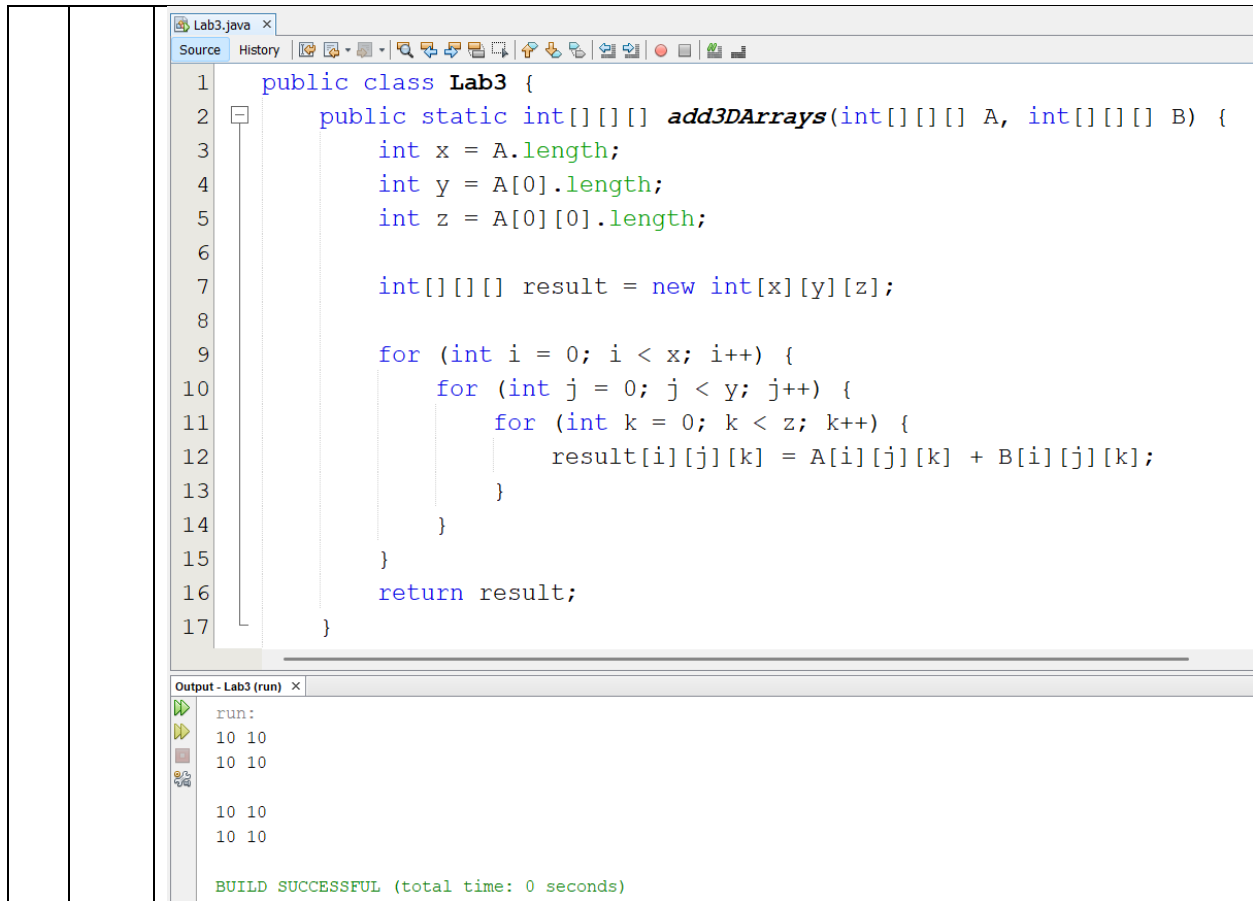
Data Structure Lab3 -Arrays

		 <pre> 1 import java.util.Random; 2 3 public class Lab3 { 4 public static void shuffle(int[] A) { 5 Random rand = new Random(); 6 for (int i = A.length - 1; i > 0; i--) { 7 int j = rand.nextInt(i + 1); // اختيار موقع عشوائي بين 8 // A[i] و A[j] تبديل 9 int temp = A[i]; 10 A[i] = A[j]; 11 A[j] = temp; 12 } 13 } 14 15 public static void main(String[] args) { 16 int[] A = {1, 2, 3, 4, 5, 6, 7, 8, 9}; // مصفوفة للاختبار 17 System.out.println("قبل الخلط"); 18 } 19 } </pre> Output - Lab3 (run) <pre> run: قبل الخلط: 1 2 3 4 5 6 7 8 9 بعد الخلط: 5 2 8 3 4 7 6 9 1 BUILD SUCCESSFUL (total time: 0 seconds) </pre>
1 3	C- 3.2 3	Suppose you are designing a multiplayer game that has $n \geq 1000$ players, numbered 1 to n, interacting in an enchanted forest. The winner of this game is the first player who can meet all the other players at least once (ties are allowed). Assuming that there is a method $\text{meet}(i, j)$, which is called each time a player i meets a player j (with $i \neq j$), describe a way to keep track of the pairs of meeting players and who is the winner.

Data Structure Lab3 -Arrays

		 <pre> 19 // التحقق من الفائز 20 if (meetings[i].size() == n - 1) { 21 System.out.println("الفائز هو اللاعب " + i); 22 } 23 if (meetings[j].size() == n - 1) { 24 System.out.println("الفائز هو اللاعب " + j); 25 } 26 } 27 28 29 public static void main(String[] args) { 30 meet(0, 1); 31 meet(0, 2); 32 meet(1, 2); 33 meet(0, 3); 34 meet(1, 3); 35 meet(2, 3); 36 } </pre> Output - Lab3 (run) x <pre> run: تم تسجيل بعض اللقاءات! BUILD SUCCESSFUL (total time: 0 seconds) </pre>
1 4	C- 3.2 4	Write a Java method that takes two three-dimensional integer arrays and adds them componentwise.

Data Structure Lab3 -Arrays



The screenshot shows an IDE window titled "Lab3.java" with a source code editor and an output console. The code defines a class `Lab3` with a static method `add3DArrays` that takes two 3D integer arrays `A` and `B` and returns a new 3D array `result` of the same dimensions. The dimensions are determined by `A.length`, `A[0].length`, and `A[0][0].length`. The method uses three nested loops to iterate over each element of `A` and `B`, adding their values to the corresponding element in `result`. The output console shows the result of running the program, displaying the dimensions of the arrays and the values of the elements in the resulting array.

```
1 public class Lab3 {  
2     public static int[][][] add3DArrays(int[][][] A, int[][][] B) {  
3         int x = A.length;  
4         int y = A[0].length;  
5         int z = A[0][0].length;  
6  
7         int[][][] result = new int[x][y][z];  
8  
9         for (int i = 0; i < x; i++) {  
10             for (int j = 0; j < y; j++) {  
11                 for (int k = 0; k < z; k++) {  
12                     result[i][j][k] = A[i][j][k] + B[i][j][k];  
13                 }  
14             }  
15         }  
16         return result;  
17     }  
}
```

Output - Lab3 (run) ×

```
run:  
10 10  
10 10  
10 10  
10 10  
10 10  
BUILD SUCCESSFUL (total time: 0 seconds)
```